

Study Questions: Perception (Fiber & Flow)

1. How do a crocheter's fingers simultaneously read multiple sensory channels (diameter, density, resistance, drape) to determine yarn weight? How is this an example of Bayesian perception?
2. Why does a knot in smooth yarn produce more surprise (prediction error) than a knot in textured yarn? What does this tell us about the role of baseline predictions in perception?
3. Describe three different types of "surprise" a crocheter might encounter while working — situations where sensory input deviates significantly from prediction.
4. How does interoception (awareness of internal bodily states) affect crochet quality? Give examples of interoceptive signals a crocheter should monitor.
5. Explain the concept of "active sensing" using the example of the squeeze test, snap test, or drape test with yarn. Why are these epistemic actions rather than pragmatic actions?
6. How does working with variegated yarn add a visual dimension to the perceptual challenge of crochet? What predictions does the crocheter form about color sequences?
7. What is multimodal perception, and how do touch and vision integrate when a crocheter works with a self-striping yarn?
8. Why does familiar yarn produce a sense of perceptual comfort? Frame your answer in terms of prediction error and the generative model.
9. A crocheter notices a thin spot in the yarn before it becomes visible — they feel it. What sensory channels detected this, and why did it register as prediction error?
10. How does the perception of yarn drape differ between a stiff cotton and a fluid silk? What predictions does each material confirm or violate?
11. Describe how color pooling in variegated yarn is a perceptual phenomenon. How does the crocheter learn to predict and manipulate pooling patterns?

12. How might ambient lighting conditions affect the perception of yarn color? Why might a crocheter choose different colors under different lighting and later be surprised?
13. Explain why the "burn test" for fiber identification is an example of actively seeking information to reduce uncertainty about hidden states.
14. How does hand fatigue change perception during a long crochet session? What happens to prediction accuracy as interoceptive signals of tiredness increase?
15. Compare the perceptual challenge of working with a smooth, solid-color yarn versus a highly textured, variegated novelty yarn. Which requires more perceptual bandwidth, and why?
16. How does the crocheter's perception of tension change as they work a project from start to finish? Does tension perception become more or less precise over the course of a project?
17. Why might two crocheters perceive the same yarn differently? How do individual differences in generative models lead to different perceptual experiences?
18. How does the sound of crochet (the click of the hook, the swish of the yarn) contribute to the crocheter's perceptual model? Is sound a meaningful sensory channel in this context?
19. Describe a situation where a crocheter's visual perception and tactile perception provide conflicting information about a yarn. How does the perceptual system resolve this conflict?
20. How does the concept of "hand feel" — the overall tactile impression of a yarn — relate to Active Inference? Is hand feel a single percept or the integration of multiple sensory streams?